

Deception Detection

Akshat Chaw Parmar

IIIT-Delhi

2022050

akshat22050@iiitd.ac.in

Rishi Pendyala

IIIT-Delhi

2022403

rishi22403@iiitd.ac.in

Vimal Jayant Subburaj

IIIT-Delhi

2022571

vimal22571@iiitd.ac.in

Abstract

Detecting deception in natural language is a challenging task, particularly in dynamic, multi-party interactions where trust and strategy intertwine. In this work, we address the problem of identifying deceptive messages within long-term, text-based conversations by modeling both linguistic cues and contextual metadata. Specifically, we develop a deception detection model for the strategic game *Diplomacy*, where players often mix truthful and false statements to manipulate outcomes. Our approach combines lexical and syntactic features with game-specific metadata such as player relationships and board state. The model is evaluated using accuracy and Macro F1, demonstrating strong performance in distinguishing deceptive from truthful communication. This study provides a foundation for improving deception detection in strategic, conversational settings.

1 Introduction

Deception is a subtle yet powerful tool in strategic communication, often blending seamlessly with truthful dialogue. Accurately identifying deception in such contexts is a complex and open problem, particularly when messages are exchanged over extended interactions with shifting alliances and goals. In this work, we explore deception detection in the context of interactive, multi-turn communication, where players must reason about trust, intent, and misdirection.

Our task is to build a model that can classify whether a given message is deceptive or truthful based on its content and context. Unlike traditional deception detection tasks that operate in isolated or static environments, our setting reflects a more realistic challenge: players engage in dynamic, goal-driven conversations that evolve over time. The presence of both sender intent and receiver perception adds further nuance, allowing us to explore

deception not only as an act but as an interpreted signal.

To tackle this, we develop a hybrid model that incorporates linguistic features—such as lexical choice, sentence structure, and sentiment—with metadata related to the game’s progress and player dynamics. We evaluate our approach using standard metrics such as accuracy and Macro F1 to measure how well it captures deceptive intent. Our results show that contextual cues, when combined with textual features, significantly enhance deception detection capabilities. This work contributes to the broader field of natural language understanding by advancing methods for identifying deception in strategically complex environments.

2 Literature Review

This section presents a comparative literature review of two notable papers on deception detection in natural language: *It Takes Two to Lie: One to Lie and One to Listen* by Peskov et al. (2020), and *The Mafiascum Dataset: A Large Text Corpus for Deception Detection* by de Ruiter and Kachergis (2019). Both papers introduce novel datasets and apply machine learning techniques to detect deception, though their contexts, data structures, and modeling approaches differ significantly.

The first study by Peskov et al. explores deception within the game of Diplomacy, a negotiation-centric, strategic board game. The dataset consists of 17,289 annotated messages collected from twelve games involving 84 players. Each message is annotated in real-time by the sender (to mark actual lies) and the receiver (to indicate suspected lies). This setup allows for a fine-grained, message-level understanding of deception as it unfolds within long-term relationships and strategic interactions. The features extracted for modeling include lexical patterns (such as bag-of-words and Harbinger categories), conversational context (pre-

vious dialogue), and social dynamics such as power differentials between players based on in-game strength.

To detect lies, the authors apply both traditional and neural approaches. Logistic regression models provide interpretable insights using linguistic cues. Additionally, long short-term memory (LSTM) models, and hierarchical LSTM architectures incorporating contextual and power features, are used for improved performance. The most effective model combines contextual LSTM with power dynamics and BERT embeddings, achieving a Lie F1-score of 27.0%, approaching human-level performance (22.5%). The results suggest that incorporating strategic and contextual information significantly improves deception detection in complex social environments.

In contrast, the second study by de Ruiter and Kachergis introduces the *Mafiascum* dataset, a large corpus of text from over 700 online forum-based Mafia games. Each document in this dataset comprises all posts by a single player in a single game, resulting in over 9,000 labeled documents. Labels are derived from players' in-game roles, categorizing Mafia members as deceptive and Town players as non-deceptive. The average document length is about 3,940 words, and documents with fewer than 50 words are excluded.

The authors explore both hand-engineered linguistic features grounded in deception literature and average word embeddings enriched with subword information (via FastText). A logistic regression model trained on a combination of these features serves as the primary classification method. The best results on documents with over 5000 words show an average precision (AP) of 0.39 and an AUROC of 0.68, compared to a chance baseline of 0.26. On shorter documents (50+ words), the AP drops to 0.29. The study confirms several previously identified linguistic cues to deception, such as increased use of negations and third-person pronouns, and emphasizes the variability of deceptive language depending on context and interactivity.

3 Dataset Description

The QANTA Diplomacy dataset is derived from full-game transcripts of the strategic negotiation game *Diplomacy*, in which seven players vie for territorial dominance through alliances, betrayals, and coordinated moves. Unlike other deception datasets which focus on static or isolated utter-

ances, this dataset reflects the longitudinal nature of human deception, embedded in evolving social relationships and strategic game states.

The dataset consists of over 17,000 messages exchanged across multiple games. Each data point represents a single message from one player to another, and includes a rich set of annotations and metadata. Key components are as follows:

- **Message Content:** Each message is a string of text written by a player to another, often involving negotiation, persuasion, or information exchange.

- **Deception Labels:**

- `sender_label`: Indicates whether the sender believes the message they sent was truthful (True) or deceptive (False).
- `receiver_label`: Indicates whether the recipient perceived the message as truthful or deceptive.

This dual-labeling enables the analysis of both intent and perception, making the dataset ideal for studying deception and miscommunication.

- **Game Metadata:**

- `season`, `year`: Track the point in the game when the message was sent (e.g., "Spring 1902").
- `game_id`: Identifies which of the multiple games the message belongs to.
- `absolute_message_index`, `relative_message_index`: Provide message order both globally and within the player-pair conversation.

- **Player Context:**

- `speaker`, `receiver`: The identities of the communicating players.
- `players`: The two players involved in a message exchange pair.

- **Game Score Metrics:**

- `game_score`: The current score or standing of the speaker at the time of message transmission.
- `game_score_delta`: The change in score from the previous turn, which can signal betrayal or cooperation dynamics.

This dataset is unique in combining message-level linguistic information with temporal, strategic, and social metadata. It allows the study of deception in the context of long-term alliances and betrayals, where truth and lies are not absolute but strategically employed. The temporal and relational annotations enable modeling how deception evolves throughout the game and how players adjust their strategies based on outcomes and trust dynamics. This makes the dataset especially powerful for building and evaluating models that aim to mimic or surpass human-level deception detection in dynamic, high-stakes environments.

4 Methodology

4.1 Exploratory Data Analysis

To better understand the structure and patterns within the QANTA Diplomacy dataset, we conducted a comprehensive exploratory data analysis (EDA). This step provided key insights that informed our preprocessing and modeling decisions.

- **Target Distribution:** We visualized the distribution of `game_score_delta`, which reflects changes in a player's game score over time. The variable shows a heavy skew with most values centered near zero, indicating that substantial changes in score are relatively rare.
- **Message Length:** Message lengths were analyzed by character count. Most messages fell within a moderate range (typically between 20 to 100 characters), suggesting players often communicate in concise, sentence-length exchanges. This helps determine padding and truncation parameters for neural models.

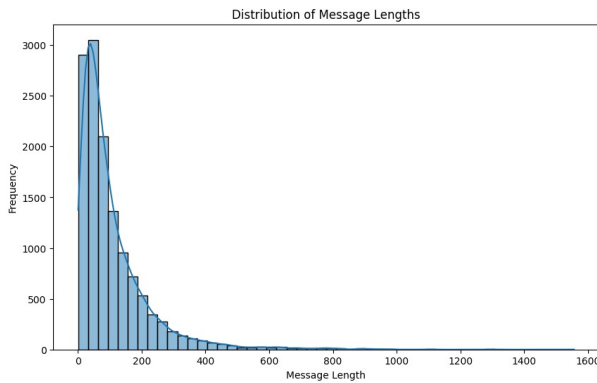


Figure 1: Distribution of message lengths across the QANTA Diplomacy dataset, highlighting variations in negotiation complexity.

- **Speaker and Receiver Frequency:** We examined which players were most active as speakers and receivers. Some player roles appeared significantly more frequently, which may introduce bias or reflect dominant game-play patterns that could influence deception.
- **Temporal Trends:** The dataset was also broken down by season and year, showing that messaging was not evenly distributed across game phases. Spring and Fall seasons showed peaks in message counts, aligning with periods of high strategic decision-making in the game.

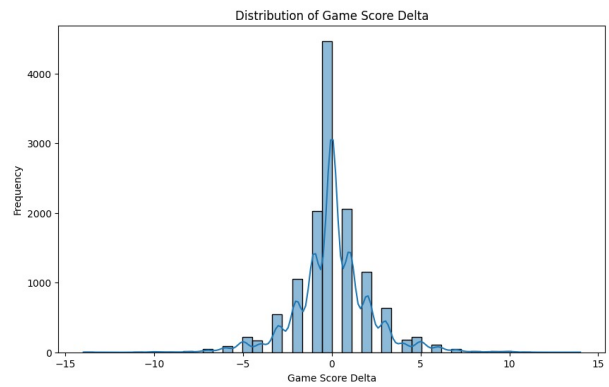


Figure 2: Temporal trends in game score deltas, reflecting strategic shifts due to alliances and betrayals.

- **Correlation Analysis:** A heatmap of numerical features (e.g., `game_score`, `game_score_delta`, `message length`, etc.) showed generally low correlation among variables. This highlights the importance of engineering new linguistic and structural features rather than relying on raw metadata alone.

Overall, the EDA reinforced that deception in Diplomacy is not tied to a single numerical factor but emerges from a combination of linguistic patterns, strategic context, and interpersonal dynamics. These findings shaped our feature extraction strategy, encouraging us to combine text-based, stylistic, and game-specific features for robust deception modeling.

4.2 Feature Extraction

This section outlines the logic behind the features used in the deception detection model, the rationale for their inclusion, and the way they are quantified to support classification.

Deceptive communication often involves subtle linguistic patterns that differ from truthful speech. These include emotional tone, sentence complexity, pronoun usage, rhetorical structuring, and shifts in formality. By quantifying these signals, we can extract meaningful indicators that correlate with deceptive intent.

4.2.1 Sentiment Polarity

Sentiment polarity reflects the emotional tone of the message. Deceptive text may lean toward emotionally charged language—either strongly positive (to appear persuasive) or negative (to distract or deflect). This feature is represented as a continuous value between -1 (very negative) and 1 (very positive), capturing emotional skewness.

4.2.2 Stylometric Features

Stylometric features capture the author’s stylistic fingerprint, many of which may shift when a person is lying. These include:

- **Average sentence length:** Longer or shorter sentences may reflect effort to obscure or embellish truth.
- **Average word length:** Simpler or more complex vocabulary may signal cognitive load.
- **Type-token ratio:** Measures vocabulary richness, which can vary with intent.
- **Function word count:** Frequencies of common words (e.g., “of”, “and”, “to”) can indicate fluency or hesitation.
- **Pronoun usage:** High or low usage of personal pronouns may reveal distancing behavior.
- **Third-person pronoun count:** Frequent references to others may indicate deflection.
- **Readability scores:** These include the Flesch Reading Ease and Flesch-Kincaid Grade Level metrics, which help assess the complexity and accessibility of text.
 - *Flesch Reading Ease:* This test rates text on a 100-point scale. The higher the score, the easier the text is to understand. For most standard documents, the recommended score falls between 60 and 70.

Formula:

$$206.835 - (1.015 \times \text{ASL}) - (84.6 \times \text{ASW})$$

where:

- * ASL = average sentence length (number of words divided by number of sentences)
 - * ASW = average number of syllables per word (number of syllables divided by number of words)
- *Flesch-Kincaid Grade Level:* This test estimates the U.S. school grade level required to understand the text. For example, a score of 8.0 suggests that an eighth grader should be able to comprehend the text. Most general-purpose documents aim for a score between 7.0 and 8.0.

Formula:

$$(0.39 \times \text{ASL}) + (11.8 \times \text{ASW}) - 15.59$$

where:

- ASL = average sentence length (number of words divided by number of sentences)
 - ASW = average number of syllables per word (number of syllables divided by number of words)
- These readability metrics help capture variation in how deceptive and truthful messages are constructed, particularly in terms of clarity, density, and word choice.
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- **Punctuation counts:** Usage of commas, periods, or exclamation points may reflect emotional or rhetorical tone.

These features are captured as counts, ratios, or normalized scores.

4.2.3 Passive Voice

Passive constructions often indicate attempts to obscure agency or responsibility. Deceivers may favor passive phrasing (e.g., “Mistakes were made” vs “I’ve made mistakes”) to avoid direct ownership. This is treated as a binary indicator: 1 if passive voice is detected, 0 otherwise.

4.2.4 Formality Score

The formality of language is a soft signal of intent. Deceptive messages may be overly formal to appear credible or excessively casual to seem innocent. This is quantified on a 0 to 1 scale, with higher scores indicating more formal tone.

4.2.5 Harbinger Word Features

Harbinger words are rhetorical cues that help structure arguments, express subjectivity, or convey logical relationships. Their presence often reflects strategic discourse. We track both the count and presence (binary) of words from categories such as claim, premise, subjectivity, and contingency. These are typically normalized by message length or used directly as raw counts.

The harbinger word set is taken from [INSERT CITATION]

4.2.6 Meta-data

Specifically for learning on the QANTA Diplomacy Dataset, we included meta-data about the game dynamics. This included temporal factors such as season and year, player information, and game state variables such as game score and game score delta.

4.2.7 Combined Use

All features are transformed into numerical representations and concatenated to form a fixed-size input vector. Together, they allow the model to detect tone, structure, behavioral markers, and situational context, contributing to significant insight when modeled together.

4.3 Model Architecture

The proposed model is a neural network that integrates a Long Short-Term Memory (LSTM) network with additional numerical features for enhanced prediction. The architecture consists of the following components:

- **Embedding Layer:** A pre-trained embedding matrix is utilized to convert input sequences into dense vector representations. The embedding layer has a vocabulary size determined by the embedding matrix and an embedding dimension of d_{emb} . This layer is frozen during training to retain the pre-trained weights.
- **LSTM Layer:** A single-layer LSTM processes the embedded sequences, capturing temporal dependencies. It takes the embedding dimension d_{emb} as input and produces a

hidden state of dimension h_{dim} . The LSTM operates in a batch-first configuration, outputting the hidden state from the final time step.

- **Feature Concatenation:** The LSTM’s final hidden state (of size h_{dim}) is concatenated with additional numerical features (of size n_{feat}), forming a combined feature vector of dimension $h_{dim} + n_{feat}$.
- **Fully Connected Layers:** The combined feature vector is passed through two fully connected layers:
 - The first layer maps the combined input ($h_{dim} + n_{feat}$) to a hidden dimension h_{dim} , introducing non-linearity.
 - The second layer maps the hidden dimension h_{dim} to the output dimension o_{dim} , producing the final prediction.

The model takes two inputs: a sequence of tokens (e.g., words or indices) and a vector of numerical features. The output is a prediction of size o_{dim} , tailored to the task at hand (e.g., classification or regression).

4.3.1 Tried Models and Failed

In addition to the final LSTM-based model, two alternative architectures were explored for deception detection on the Qanta dataset, which is derived from the Diplomacy game and characterized by severe class imbalance. These models, described below, did not perform satisfactorily, likely due to their architectural limitations in handling the task’s complexity and the dataset’s imbalance.

- **Transformer-Based Model:** This model replaces the LSTM with a Transformer encoder to process the input sequences. Its architecture comprises:
 - **Embedding Layer:** Identical to the final model, it uses a frozen pre-trained embedding matrix to map input tokens to vectors of dimension d_{emb} .
 - **Transformer Encoder:** A two-layer Transformer encoder with five attention heads processes the embedded sequences. Each layer has a feedforward dimension of h_{dim} . The input to the Transformer is permuted to shape (seq_len, batch_size, d_{emb}), and the output, also

of shape $(seq_len, batch_size, d_{emb})$, is averaged over the sequence dimension to yield a representation of shape $(batch_size, d_{emb})$.

- **Feature Concatenation:** The averaged Transformer output is concatenated with numerical features (n_{feat}) , forming a vector of size $d_{emb} + n_{feat}$.
- **Fully Connected Layers:** Two linear layers map the concatenated vector first to h_{dim} and then to o_{dim} , producing the final prediction.

Reasons for Failure: The Transformer model may have underperformed due to several factors. Transformers are highly effective for capturing long-range dependencies but require substantial data to generalize well, especially with multiple attention heads and layers. The Qanta dataset’s limited size and severe class imbalance likely hindered the model’s ability to learn robust patterns, leading to overfitting on the majority class. Additionally, averaging the Transformer output over the sequence dimension may have discarded critical temporal or positional information essential for detecting deceptive patterns in Diplomacy game dialogues.

- **Simple Feedforward Model:** This model adopts a minimalist approach, bypassing recurrent or attention-based processing. Its architecture includes:

- **Embedding Layer:** As in the other models, a frozen pre-trained embedding matrix converts tokens to vectors of dimension d_{emb} . The embedded sequence is averaged over the sequence dimension to produce a single vector of shape $(batch_size, d_{emb})$.
- **Feature Concatenation:** The averaged embedding is concatenated with numerical features (n_{feat}) , resulting in a vector of size $d_{emb} + n_{feat}$.
- **Fully Connected Layers:** Two linear layers process the concatenated vector, mapping it to h_{dim} and then to o_{dim} for the final prediction.

Reasons for Failure: The simplicity of this model likely contributed to its poor performance. By averaging the embedded sequence,

the model discards all temporal and contextual relationships within the input sequence, which are critical for deception detection in the Diplomacy game, where the order and context of statements influence meaning. This lack of sequence modeling capability made it difficult for the model to capture subtle cues of deception. Furthermore, the severe class imbalance in the Qanta dataset likely exacerbated the model’s tendency to overfit to the majority class, as the feedforward architecture lacks the capacity to learn complex patterns that might distinguish rare deceptive instances.

Both models struggled to address the challenges posed by the Qanta dataset’s imbalance and the nuanced nature of deception detection. The final LSTM-based model succeeded where these models failed, likely due to its ability to effectively model sequential dependencies while integrating numerical features, providing a balanced approach to capturing both temporal and contextual information.

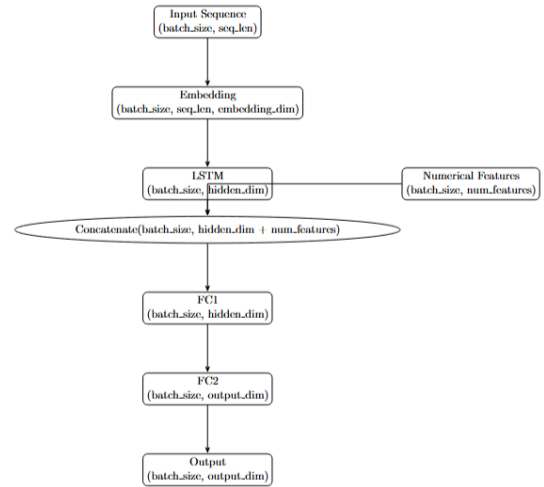


Figure 3: Final model Architecture

5 Experiments and Results

5.1 Evaluation Metrics

We report accuracy and macro-F1. We are also reporting the Lie F1 as we have found having a high macro F1 doesn't necessarily do well in the Lie category due to the data imbalance.

5.2 Results

Our model gave the following results on the test dataset:

Model	Accuracy	Macro F1	Lie F1
Baseline 1 (LSTM)	90.84	49.51	-
Baseline 2 (LR)	91	.48	-
Paper Model	N.A.	.57	.27
Final Model	88	0.60	0.26

Table 1: Comparison of model performance across baselines and proposed models.

6 Conclusion

We demonstrate that a multi-modal approach to deception detection provides significant improvements over single modality methods. Future work includes incorporating contextual cues and exploring transformer-based fusion mechanisms. Our Github Repo Link : [GITHUB](#)

References

- [1] Denis Peskov, Benny Cheng, Ahmed Elgohary, Joe Barrow, Cristian Danescu-Niculescu-Mizil, Jordan Boyd-Graber. 2020. *It Takes Two to Lie: One to Lie, and One to Listen*. University of Maryland, Cornell University.
- [2] Sheila Mms. Diplomacy-NLP. *GitHub Repository*. Available: <https://github.com/shielamms/Diplomacy-NLP>
- [3] Denis Peskov. 2020 ACL Diplomacy. *GitHub Repository*. Available: https://github.com/DenisPeskov/2020_acl_diplomacy/tree/master
- [4] Denis Peskov, Benny Cheng, Ahmed Elgohary, Joe Barrow, Cristian Danescu-Niculescu-Mizil, Jordan Boyd-Graber. 2020. *It Takes Two to Lie: One to Lie and One to Listen*. Proceedings of the Association for Computational Linguistics. Available: http://cs.umd.edu/~jbg/docs/2020_acl_diplomacy.pdf
- [5] Bob de Ruiter and George Kachergis. 2019. *The Mafiascum Dataset: A Large Text Corpus for Deception Detection*. arXiv preprint arXiv:1811.07851v3. Available: <https://arxiv.org/abs/1811.07851>